

1 Highlights

2 **Water transparency retrieval in a high-altitude large lake from** 3 **Sentinel-2 and Landsat 8/9: cross-sensor harmonization limits, val-** 4 **idation design, and calibrated uncertainty in Lake Titicaca**

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- 6 • Land-derived Landsat/Sentinel-2 harmonization is invalid over dark
7 oligotrophic water
- 8 • Sensors stay 99.8% separable after harmonization; intercepts exceed
9 signal 70-fold
- 10 • Campaign, not station, is the blocking unit in campaign-based moni-
11 toring programmes
- 12 • Honest Secchi retrieval: $R^2=0.57$, RMSE 1.86 m over 812 real match-
13 ups
- 14 • Conformal intervals are calibrated marginally but under-cover in the
15 clearest water

16 Water transparency retrieval in a high-altitude large
17 lake from Sentinel-2 and Landsat 8/9: cross-sensor
18 harmonization limits, validation design, and calibrated
19 uncertainty in Lake Titicaca

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22 **Abstract**

23 Monitoring water transparency in high-altitude oligotrophic lakes is critical
24 for freshwater management, yet it remains constrained by the sparsity of field
25 data. While multi-mission satellite archives (e.g., harmonized Sentinel-2 and
26 Landsat 8/9) and machine learning offer a synoptic solution, current practices
27 often import assumptions from turbid-water literature that fail in clear-water,
28 campaign-based settings. We assemble 1002 satellite–field match-ups against
29 twelve years of limnological monitoring in Lake Titicaca to address these fail-
30 ures and define a defensible retrieval protocol. First, we demonstrate that
31 standard land-derived harmonization coefficients (e.g., Roy et al., 2016) are
32 invalid over dark water, as their intercepts exceed the native Landsat signal
33 by factors of 1.4 to 69.8, leaving sensors 99.8% separable; we propose a clas-
34 sifier two-sample test to catch this failure. Second, we show that standard
35 station-blocked cross-validation fails to prevent spatial leakage in campaign-
36 based sampling: under campaign-blocked folds, a station-mean null model
37 still explains $R^2 = 0.208$ of the variance. We introduce a null-model diag-

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nostic to identify the correct blocking unit (the campaign). Finally, under this strict design, we train a Random Forest regressor to retrieve Secchi disk depth ($R^2 = 0.574$, RMSE 1.86 m) and attach split-conformal prediction intervals. We reveal that while marginally calibrated (89.4% coverage at 90% nominal), conformal intervals structurally under-cover in the clearest water (71.3%). Our findings provide a rigorous blueprint for evaluating machine-learning water quality products in large, oligotrophic lakes.

Keywords: Lake Titicaca, Secchi disk depth, Sentinel-2, Landsat, cross-sensor harmonization, conformal prediction, large lakes

1. Introduction

Large lakes are sentinels of environmental change, but continuous monitoring of remote, high-altitude systems is constrained by sparse field infrastructure (Williamson et al., 2009). Lake Titicaca (3,810 m a.s.l.), the largest tropical lake in the world by volume, is a critical transboundary freshwater reserve whose water quality is documented only by infrequent field campaigns (OEFA, 2016; Siguayro Mamani et al., 2022), even as eutrophication advances in its near-shore bays (Heredia et al., 2022; Lanza et al., 2024). Satellite remote sensing offers a synoptic solution (Gholizadeh et al., 2016; Topp et al., 2020). Specifically, water transparency—quantified by the Secchi disk depth (Z_{SD})—is an integrative indicator of water clarity strongly linked to light attenuation (Lee et al., 2015). Because Sentinel-2 MSI and Landsat 8/9 OLI provide complementary revisit times, their joint use as a virtual constellation coupled with machine learning is now the standard practice for mapping water quality (Pahlevan et al., 2020; Cao et al., 2022).

62 Although these virtual constellations work well in turbid, productive
 63 “Case-2” waters, they fail to transfer reliably to clear, oligotrophic high-
 64 altitude lakes due to two unresolved methodological assumptions. First,
 65 the cross-sensor harmonization coefficients required to merge Landsat and
 66 Sentinel-2 (e.g., Roy et al., 2016) were fitted on terrestrial targets. Because
 67 terrestrial surfaces are orders of magnitude brighter than clear water, their
 68 additive intercepts overwhelm the over-water signal. While near-infrared
 69 bias over water has been noted (Pahlevan et al., 2019), the magnitude of
 70 this mismatch and its effect on machine-learning retrievals remain unquan-
 71 tified. Second, to prevent spatial autocorrelation from inflating apparent
 72 accuracy, previous methods routinely block cross-validation by monitoring
 73 station (Roberts et al., 2017). However, they assume rather than test that
 74 station identity is the primary source of data leakage. In campaign-based
 75 monitoring programmes, where an entire lake is sampled within a few days
 76 under identical conditions, station-blocking may retain severe leakage and
 77 falsely inflate performance metrics.

78 In this paper, we propose a rigorous validation and retrieval pipeline to
 79 overcome these challenges, using Lake Titicaca as our testing ground. Our
 80 innovation is replacing assumed methodological defaults with data-driven
 81 diagnostics. Specifically, we assemble 1002 real satellite–field match-ups and
 82 apply a classifier two-sample test (adversarial validation) (Lopez-Paz and
 83 Oquab, 2017) to diagnose harmonization failure. Furthermore, we introduce
 84 a null-model diagnostic to identify the true blocking unit for cross-validation.
 85 Finally, we train a Random Forest regressor under the corrected design and
 86 attach split-conformal prediction intervals (Angelopoulos and Bates, 2023)

87 to quantify per-prediction uncertainty.

88 In contrast to previous water-quality mapping efforts (Ayala Izurieta
89 et al., 2023; Maligaya et al., 2024), our study makes three explicit contri-
90 butions:

- 91 **1. Quantification of Harmonization Limits:** We show that land-
92 derived harmonization intercepts exceed the native Landsat signal by
93 up to 69.8-fold over clear water. Consequently, a classifier can identify
94 the source sensor with 99.8% accuracy from harmonized bands, proving
95 that the model learns a sensor label rather than unified optics.
- 96 **2. Null-Model Validation Design:** We demonstrate that under campaign-
97 blocked folds, a station-mean null model still achieves $R^2 = 0.208$,
98 proving that station-blocking fails to prevent leakage. We establish the
99 campaign as the required blocking unit for this class of data.
- 100 **3. Defensible Retrieval with Conformal Uncertainty:** Under this
101 strict design, our model achieves an honest R^2 of 0.574 (RMSE 1.86 m).
102 We further reveal that while split-conformal intervals are marginally
103 calibrated (89.4% coverage), they structurally under-cover extreme val-
104 ues in the clearest water, highlighting a critical limitation for opera-
105 tional monitoring.

106 2. Materials and methods

107 2.1. Study area

108 Lake Titicaca comprises three limnologically distinct zones (Fig. 1): the
109 large, deep, oligotrophic Lago Mayor; the shallower Lago Menor (Wiñay-
110 marca) to the south-east; and Bahía de Puno, which receives municipal and

111 agricultural inputs from the city of Puno and is the most eutrophic sector.
112 Measured transparency spans 1.5 m in near-shore turbid water to 16.5 m in
113 the open oligotrophic basin.

114 2.2. *In-situ data*

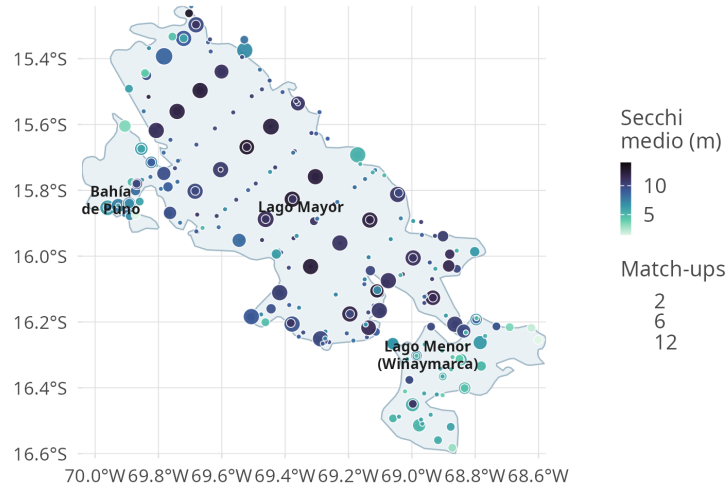
115 We used 881 field measurements of Secchi transparency, with co-measured
116 chlorophyll-*a*, suspended solids, temperature, pH, dissolved oxygen and nu-
117 trients, from the limnological monitoring campaigns of IMARPE (Instituto
118 del Mar del Perú) on Lake Titicaca, spanning 2011–2024 across 156 sta-
119 tions distributed over the whole lake. Of these, 734 include a valid Secchi
120 reading. The July 2019 campaign (71 stations) is published by Siguayro Ma-
121 mani and Franco (2022); the remaining years derive from the same IMARPE
122 programme. As a provenance check, our 2019 zone-level means reproduce
123 the published values almost exactly (Lago Mayor conductivity 1513.4 vs.
124 1513.5 $\mu\text{S cm}^{-1}$; Lago Menor chlorophyll-*a* 1.52 vs. 1.52 mg m^{-3}).

125 Three properties of this record constrain everything that follows, and we
126 state them here rather than in the limitations (Fig. A.5).

127 First, the monitoring programme is itself almost entirely a dry-season
128 programme. The 2011–2024 record contains 13 campaigns with Secchi read-
129 ings, of which exactly one — December 2012, 72 measurements — falls in the
130 austral wet season (December–March). That campaign predates both sen-
131 sors used here: Landsat 8 became operational in May 2013 and Sentinel-2A
132 in June 2015. Consequently every match-up in our dataset falls between July
133 and October, and this is a property of the field programme rather than of
134 our processing: no choice of match-up window or cloud threshold can recover
135 wet-season pairs that have no contemporaneous imagery. We return to what

(a) Red de monitoreo y gradiente de transparencia

Estaciones IMARPE/ALT con match-up satélite-campo (2013–2024), coloreadas por transparencia media medida. El gradiente trófico es visible en el propio dato de c



(b) El esfuerzo de muestreo es muy desigual

Cada punto es una estación. 319 estaciones, pero la mayoría se visitó en muy pocas campañas, lo que limita el poder de cualquier análisis por estación

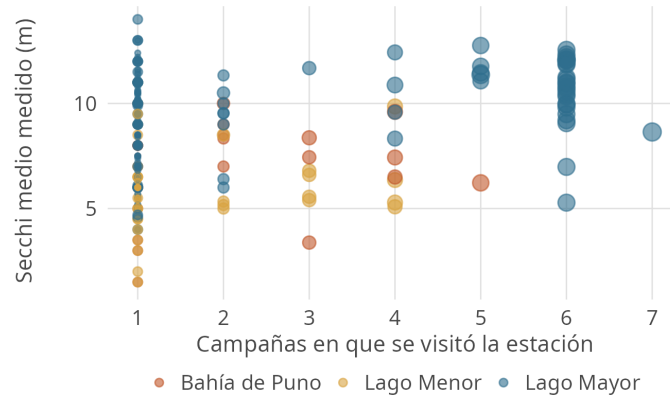


Figure 1: Study area and monitoring network. (a) IMARPE/ALT stations with a satellite–field match-up, coloured by mean measured Secchi depth over the Natural Earth lake boundary; the trophic gradient is evident in the field data alone. (b) Sampling effort per station: most stations were visited in few campaigns, which limits the power of any station-level analysis.

136 this means for operational use in Section 4.3.

137 Second, sampling is episodic: campaigns exist for 2013–2024 with no
138 campaigns in 2020, 2021, 2023.

139 Third, the reference measurement is coarse: 73 distinct Secchi values
140 occur in 812 match-ups, because disk readings are recorded to the nearest
141 half metre.

142 No synthetic or spectrally derived labels were used at any stage.

143 *2.3. Satellite data and cross-sensor harmonization*

144 Sentinel-2 MSI (harmonized surface reflectance) and Landsat 8/9 OLI
145 (Collection 2, Level-2; Vermote et al., 2016) were accessed through Google
146 Earth Engine (Gorelick et al., 2017). Clouds were masked with the Sentinel-
147 2 Scene Classification Layer plus cloud probability, and with the Landsat
148 QA_PIXEL bitmask (dilated cloud, cloud and shadow). The cirrus flag was
149 not applied, as it is near-permanently set at 3810 m and would mask the
150 entire archive. We retained the six bands common to both sensors (blue,
151 green, red, NIR, SWIR1, SWIR2).

152 Landsat surface reflectance was first harmonized to Sentinel-2 equivalent
153 reflectance with the per-band linear coefficients of Roy et al. (2016), of the
154 form $\rho_{\text{MSI}} = a + b \rho_{\text{OLI}}$. Section 3.1 evaluates whether this transform is valid
155 over the water of Lake Titicaca and derives an alternative. For that evalu-
156 ation we identified every in-situ event observed by both sensors within the
157 match-up window — 270 paired station–date observations — and regressed
158 Sentinel-2 on harmonized Landsat reflectance band by band.

159 2.4. *Satellite-field match-ups*

160 For every in-situ measurement we extracted the cloud-masked median
161 reflectance within a 30 m buffer of the station coordinate, from imagery
162 acquired within ± 10 days. The 30 m buffer matches the coarser Landsat pixel
163 and averages residual geolocation error and sensor noise over several pixels
164 while remaining in open water, away from shoreline mixing. The ± 10 -day
165 window balances the scarce cloud-free image availability at 3810 m against
166 the slow temporal change of transparency in this large, deep lake, and is
167 consistent with windows used in operational Z_{SD} retrieval (Kloiber et al.,
168 2002; Zhang et al., 2022).

169 This yielded 1002 match-ups (340 Sentinel-2, 662 Landsat), of which 812
170 have a Secchi value, across 143 stations and 120 campaign dates. Because
171 both sensors can observe the same field event, 272 match-ups are cross-sensor
172 pairs of a single in-situ measurement; collapsing them leaves 540 independent
173 events, and Section 3.2 verifies that accuracy is unchanged, confirming the
174 result is not an artefact of pseudo-replication.

175 2.5. *Features and model*

176 Twelve sensor-agnostic features were computed from the six common
177 bands: the bands themselves plus NDWI, NDTI and the ratios blue/green,
178 green/blue, red/green and blue/red. We trained a Random Forest regres-
179 sor (Breiman, 2001). Reflectance was scaled with a robust (median/IQR)
180 scaler fitted only on the training partition of each fold. All models were
181 implemented with scikit-learn (Pedregosa et al., 2011).

182 Model choice was settled empirically rather than assumed. We com-
183 pared Random Forest, Extremely Randomized Trees, XGBoost (Chen and

184 Guestrin, 2016) and LightGBM, each with hyperparameters optimized by
 185 Optuna (Akiba et al., 2019) inside *nested* cross-validation — tuning on an
 186 inner campaign-blocked split of each outer training partition, so no outer fold
 187 ever informs its own hyperparameters — and benchmarked the resulting gain
 188 against the spread across random seeds. Section 3.4 reports the outcome.

189 2.6. Validation design and null models

190 Rather than a single split, we evaluate a hierarchy of increasingly de-
 191 manding designs (Roberts et al., 2017): random K -fold; GroupKFold blocked
 192 by station; GroupKFold blocked by *campaign date*; GroupKFold blocked by
 193 year; a temporal hold-out; and leave-one-zone-out, in which an entire trophic
 194 zone is withheld.

195 Crucially, we do not assume which blocking unit matters. We benchmark
 196 each design against null models that predict, respectively, the training-set
 197 mean of the zone, of the station and of the campaign date. A blocking unit
 198 is doing real work if, and only if, the corresponding null model collapses
 199 under it. This diagnostic, rather than convention, determines our primary
 200 design.

201 Performance is reported as out-of-fold R^2 , RMSE and MAE, and bench-
 202 marked against a classical log-linear blue/green band-ratio algorithm in the
 203 standard form $\ln Z_{SD} \sim \ln(\rho_{\text{blue}}/\rho_{\text{green}})$ (Kloiber et al., 2002), with predic-
 204 tions bounded to the physically observable range, and against a multiband
 205 linear model.

206 2.7. Uncertainty and interpretation

207 We attach a per-prediction interval using split-conformal prediction (An-
208 gelopoulos and Bates, 2023): within each fold, a calibration subset of the
209 training partition yields the $(1 - \alpha)$ quantile of absolute residuals, defining
210 a symmetric interval with finite-sample marginal coverage guarantees (Ro-
211 mano et al., 2019). Because a single nominal level can be met by chance, we
212 report empirical coverage at 50, 80, 90 and 95%, and also coverage condi-
213 tional on measured transparency, since marginal calibration does not imply
214 conditional calibration.

215 The model is interpreted with SHAP (Lundberg and Lee, 2017; Lundberg
216 et al., 2020), computed *out of fold* — each observation explained by the model
217 that did not see it — rather than on a model fitted to the whole dataset, which
218 inflates the apparent importance of memorized features. Trends in annual
219 median in-situ transparency were tested with Mann–Kendall and Sen’s slope
220 (Mann, 1945; Sen, 1968) on the full 2011–2024 in-situ record, not on the
221 match-up subset.

222 3. Results

223 3.1. Land-derived harmonization is invalid over oligotrophic water

224 The Roy et al. (2016) transforms carry additive intercepts of 0.0116–
225 0.0448 in reflectance units. Over Lake Titicaca the native Landsat over-water
226 signal is far smaller than this (Table A.2, Fig. 2a). In the near-infrared the
227 median native reflectance is effectively zero, so the intercept exceeds the
228 signal by a factor of 69.8; in SWIR1 the factor is 23.2, in red 10.6, and even
229 in blue — the brightest water band — 2.5. The harmonized Landsat bands

230 are consequently dominated by a constant inherited from a land calibration:
231 their medians exceed the corresponding Sentinel-2 medians by 1.6-fold in
232 green and 15.4-fold in the near-infrared.

233 A harmonization that succeeded would make the two sensors statistically
234 indistinguishable over the same water. They are not. A Random Forest clas-
235 sifier trained to identify the source sensor from the harmonized features alone
236 reaches 99.8% accuracy against a 66.3% majority-class baseline (Fig. 2b); re-
237 stricting it to the visible bands gives 97.9%, and to scale-invariant band ratios
238 99.5%. This is a classifier two-sample test (Friedman, 2003; Lopez-Paz and
239 Oquab, 2017): if two samples come from the same distribution, no classifier
240 can separate them above chance. Sensor identity therefore remains fully en-
241 coded in the harmonized reflectance, and any model trained on the pooled
242 archive can read it. The test is not new, but to our knowledge it has not
243 been used as a harmonization acceptance check; it costs one model fit and it
244 is decisive.

245 Refitting locally on the 270 station–date events observed by both sensors
246 gives water-specific coefficients (Table A.3). These restore the spectral sig-
247 nature almost exactly — the median Landsat/Sentinel-2 ratio falls from 15.4
248 to about 1.0 across all six bands (Fig. 2d) — but per-observation agreement
249 remains uneven (Fig. 2c). Green recovers usable agreement ($r = 0.71$), and
250 NIR and SWIR reach $r \approx 0.6$. Blue does not: $r = 0.26$. This is the most
251 consequential result of the section, because SHAP attributes 72.6% of the
252 retrieval to green/blue bands and ratios (Section 3.7). The band on which
253 the retrieval most depends is the band on which the two sensors least agree.

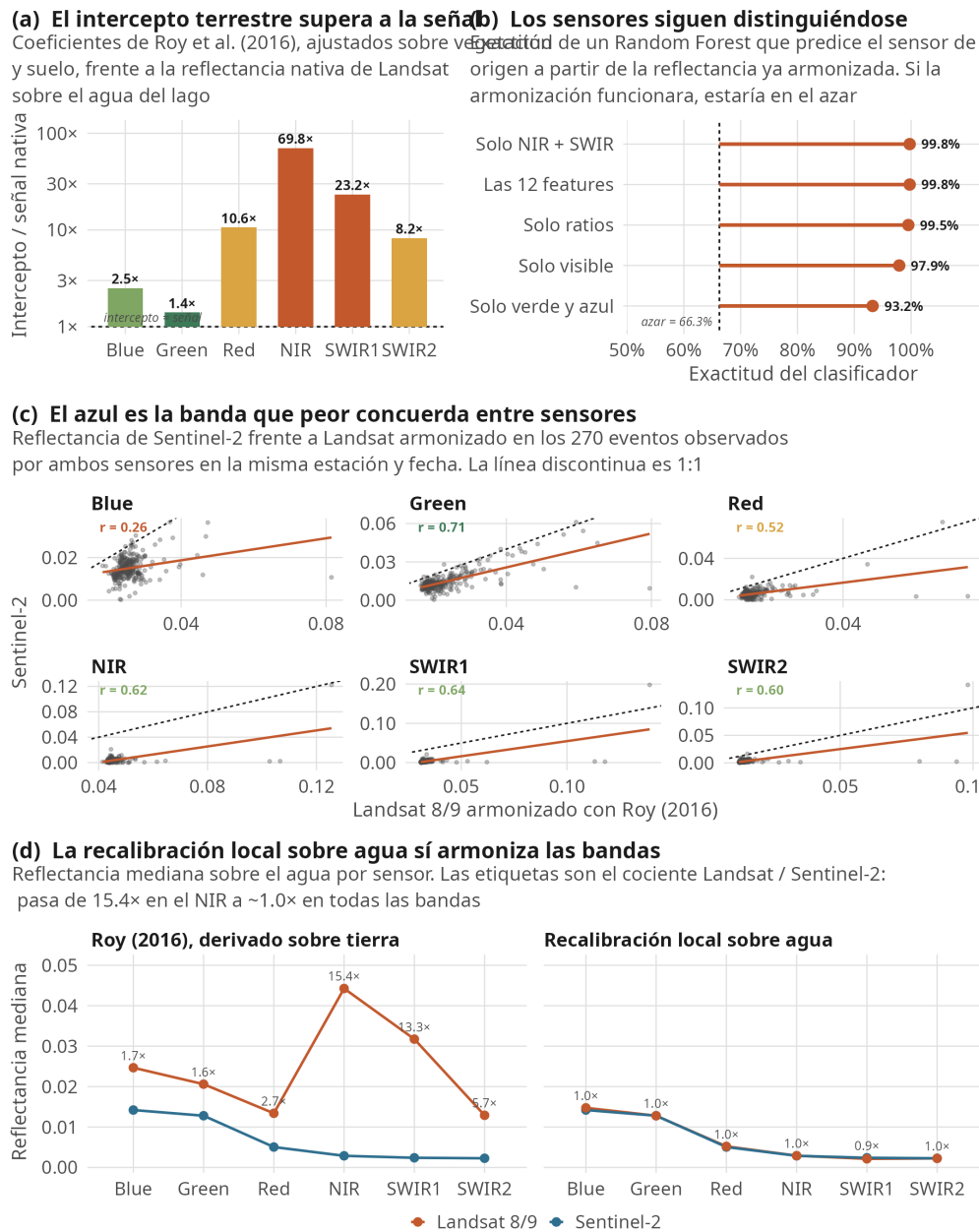


Figure 2: Why land-derived harmonization fails over clear water. (a) The Roy intercept relative to the native over-water Landsat signal, by band. (b) Accuracy of a classifier predicting the source sensor from harmonized features; under a successful harmonization it would sit at chance. (c) Sentinel-2 versus harmonized Landsat reflectance on the 270 paired station-date events; blue agrees worst. (d) Median spectral signature before and after local water recalibration; labels are the Landsat/Sentinel-2 ratio.

254 *3.2. The campaign, not the station, is the blocking unit*

255 Table 1 reports the retrieval under each validation design alongside the
256 null models (Fig. A.6). Blocking by station changes nothing relative to a ran-
257 dom split (both $R^2 = 0.596$), which under the usual reading would be taken
258 as evidence of no spatial leakage. The null models show why that reading
259 is wrong: under campaign-blocked folds the station-mean null still attains
260 $R^2 = 0.208$ and the zone-mean null $R^2 = 0.287$, whereas the campaign-date-
261 mean null collapses to -0.012 . Dependence in this dataset lives between
262 campaigns, not between stations, because a single campaign samples the
263 whole lake within a few days under one set of atmospheric and limnological
264 conditions. Station-blocked folds leave that structure intact.

265 Under campaign-blocked validation — our primary design — the Random
266 Forest attains $R^2 = 0.574$ and RMSE 1.86 m over 812 match-ups, essentially
267 unbiased (-0.02 m). Blocking by year is stricter still ($R^2 = 0.495$). Collaps-
268 ing cross-sensor pseudo-replicates to 540 independent events leaves accuracy
269 unchanged ($R^2 = 0.571$).

270 *3.3. Retrieval accuracy and benchmarks*

271 Against this design the classical blue/green band ratio attains $R^2 = 0.097$
272 (RMSE 2.71 m) and a multiband linear model $R^2 = 0.323$ (RMSE 2.34 m)
273 (Table A.4, Fig. 3). We report one methodological caution alongside these
274 figures. An unbounded implementation of the same band-ratio algorithm —
275 log-transformed and back-transformed without constraining predictions to
276 the observable range — produces a single prediction of 143.5 m against an
277 observed maximum of 16.5 m, which alone drives R^2 to -2.458 and RMSE to

Table 1: Validation design, null models and blocking units (Random Forest, $n = 812$ unless noted). A blocking unit is effective only if the corresponding null model collapses under it.

	Setting	R^2	RMSE (m)
Null models	Zone-mean	0.287	2.40
	Station-mean	0.208	2.53
	Campaign-date-mean	-0.012	2.86
Blocking unit	Random K -fold	0.596	1.81
	By station	0.596	1.81
	By campaign date	0.574	1.86
	By year	0.495	2.02
	One record per event ($n = 540$)	0.571	1.77
Zone extrapolation	Leave out Lago Menor ($n = 133$)	0.247	1.90
	Leave out Bahía de Puno ($n = 88$)	0.036	2.11
	Leave out Lago Mayor ($n = 591$)	-0.047	2.52
Temporal	Train ≤ 2019 , test 2022 & 2024 ($n = 226$)	0.507	2.63

278 5.29 m (Fig. 3d). Benchmarks of classical algorithms reported without phys-
279 ical bounds should be treated with caution: a single unconstrained extrapo-
280 lation manufactures an arbitrarily large advantage for the machine-learning
281 model.

282 3.4. *Model family and tuning do not change the picture*

283 We compared four learner families, each tuned by Optuna inside nested
284 cross-validation, against target transformations and reduced feature sets (Ta-
285 ble A.5). The best configuration — an equal-weight ensemble of the four
286 tuned models — reached $R^2 = 0.583$, an improvement of 0.009 over the
287 untuned Random Forest. That gain does exceed the spread across random
288 seeds (0.003 in R^2), so it is not noise; but it corresponds to 0.02 m of RMSE
289 on a lake whose reference measurement is recorded to the nearest half metre,
290 and it changes no conclusion in this article. Neither log-transforming the
291 target nor tuning individual learners helped more than that, and restricting
292 the feature set cost accuracy.

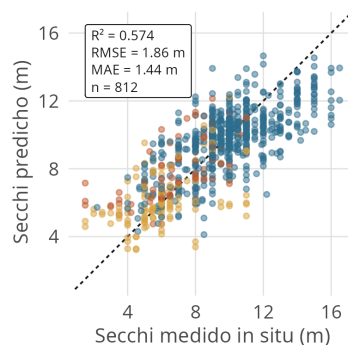
293 We therefore retain the untuned Random Forest as the reported model:
294 it is simpler, faster and easier to reproduce, and reporting a tuned ensemble
295 for a 0.009 gain would overstate the precision of the product. The ceiling
296 here is set by the data — a dry-season-only record, a reference measurement
297 quantized at half a metre, and a target whose variance is dominated by a
298 three-level trophic gradient — not by the choice of learner.

299 3.5. *Where the retrieval stops working*

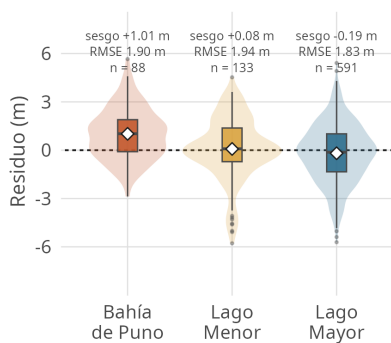
300 Withholding an entire trophic zone collapses the retrieval (Table 1): R^2
301 falls to 0.247 for Lago Menor, 0.036 for Bahía de Puno and -0.047 for Lago

(a) Predicho frente a medido

Random Forest, validación bloqueada por campaña. La discontinua es 1:1

**(b) Sobreestima donde más importa**

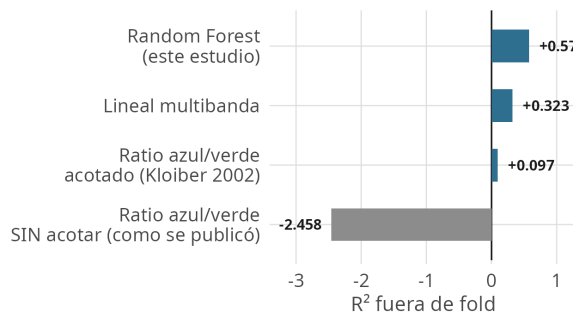
Residuos (predicho – medido) por zona. El rombo es la media. Bahía de Puno, la zona más eutrófica, +1.0 m



• Bahía de Puno • Lago Menor • Lago Mayor

(c) El baseline estaba mal implementado

Mismo diseño de validación para todos. El baseline «publicado» deshacía el log sin acotar; una única predicción de 143 m hundía su R^2

**(d) Una predicción imposible basta**

Ratio clásico en escala logarítmica. El punto de 143 m es el que producía el $R^2 = -2.46$

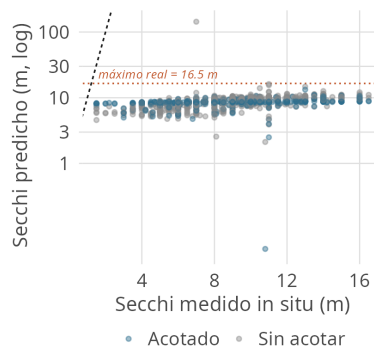


Figure 3: Retrieval under the primary design. (a) Out-of-fold predicted versus measured Secchi depth. (b) Residuals by trophic zone; the retrieval overestimates in eutrophic Bahía de Puno. (c) Benchmark against classical and linear baselines. (d) The unbounded classical implementation in log scale: one impossible prediction destroys the metric.

302 Mayor. The model interpolates within optical regimes it has seen and does
 303 not extrapolate to a water type absent from training. On the temporal hold-
 304 out (train ≤ 2019 , test 2022 and 2024) it retains $R^2 = 0.507$, though 57 of the
 305 69 test stations also appear in training (82.6%), so this measures temporal
 306 rather than fully independent generalization.

307 Error is unevenly distributed (Table A.6). The retrieval carries a positive
 308 bias of 1.01 m in Bahía de Puno, the most eutrophic and most vulnerable
 309 zone, because the clear-water majority pulls predictions upward.

310 Two features of the harmonized archive should be read together with these
 311 numbers. Removing the two bands most corrupted by the land-derived in-
 312 tercepts lowers R^2 only from 0.574 to 0.541; using scale-invariant ratios alone
 313 gives 0.468; and applying our water-specific recalibration gives 0.518. Train-
 314 ing on each sensor separately gives 0.563 (Sentinel-2, $n = 274$) and 0.549
 315 (Landsat, $n = 538$). The pooled model’s small advantage over these vari-
 316 ants is consistent with it exploiting sensor identity as an additional covariate
 317 rather than with genuine cross-sensor synergy.

318 3.6. Calibrated per-prediction uncertainty

319 Split-conformal intervals are well calibrated *marginally* at every nominal
 320 level tested: empirical coverage is 50.9% at 50% nominal, 79.1% at 80%,
 321 89.4% at 90% and 94.3% at 95%, a maximum calibration error of 0.94 per-
 322 centage points (Fig. A.7a).

323 They are not calibrated *conditionally*, and this matters more for opera-
 324 tional use than the marginal result. Coverage falls to 71.3% in the clearest
 325 water ($Z_{SD} > 12.5$ m) and 79.5% in the most turbid ($Z_{SD} < 5$ m), while
 326 exceeding the nominal level in the middle of the range (Fig. A.7c). The

327 mechanism is structural rather than incidental: split conformal derives a sin-
 328 gle global quantile of absolute residuals, so the interval has constant width
 329 (6.12 m at 90%) everywhere and cannot adapt to the local difficulty of the re-
 330 trieval. It therefore buys its marginal guarantee by over-covering in the well-
 331 sampled middle of the range and under-covering at both extremes — which
 332 are precisely where management decisions are made. A locally adaptive vari-
 333 ant such as conformalized quantile regression (Romano et al., 2019) would
 334 be required for reliable conditional coverage, and we recommend reporting
 335 coverage conditional on the predicted value as standard practice whenever
 336 conformal intervals accompany an operational product.

337 For a lake whose transparency spans 1.5–16.5 m, a ± 3 m interval is in-
 338 formative for zone-scale contrasts and multi-metre change, and is not infor-
 339 mative for sub-metre differences.

340 *3.7. Interpretation and retrievability*

341 Out-of-fold SHAP attributes 72.6% of the retrieval to green and blue
 342 bands and their ratios, 14.4% to NIR and SWIR, and the remainder to
 343 other indices (Fig. 4a). Green/blue alone accounts for 37%. The dependence
 344 is monotone and of the sign expected from water-clarity optics (Fig. 4b):
 345 a higher green-to-blue ratio pushes the prediction toward less transparent
 346 water. The retrieval is therefore grounded in optics rather than location
 347 memorization — while resting on the band pair with the worst cross-sensor
 348 agreement.

349 With real labels and the same campaign-blocked design, no other vari-
 350 able is retrievable (Fig. 4c): chlorophyll-*a* reaches $R^2 = 0.012$, suspended
 351 solids 0.056, in-situ water temperature 0.113, and Landsat thermal surface

352 temperature 0.020 ($r = 0.14$, $n = 662$). Their in-situ ranges — chlorophyll-*a*
353 0–4.6 mg m⁻³, suspended solids mostly below 10 mg L⁻¹ — lie below the op-
354 tical and thermal detection limits of these sensors. We subject the negative
355 claim to the same design as the positive one, and report it to caution against
356 applying chlorophyll algorithms (Gitelson et al., 2009; Mishra and Mishra,
357 2012) or thermal retrievals (Ruiz-Verdú et al., 2016) in clear high-altitude
358 water without local validation.

359 *3.8. Temporal patterns and the fragility of trend claims*

360 Mann–Kendall tests on annual median in-situ transparency over the full
361 2011–2024 record show no statistically significant trend in any zone (Bahía
362 de Puno: Sen slope 0.16 m yr⁻¹, $p = 0.241$; Lago Mayor: 0.25 m yr⁻¹,
363 $p = 0.138$; Fig. A.8a).

364 That conclusion is fragile in a way worth reporting explicitly. Restricting
365 the series to 2013 onwards — the natural choice if trends were computed
366 on the satellite match-up subset, whose record begins with Landsat 8 —
367 turns both Bahía de Puno ($p = 0.017$) and Lago Mayor (Sen +0.42 m yr⁻¹,
368 $p = 0.012$) significantly positive. Across start years from 2011 to 2014, 5 of
369 12 zone–start-year combinations change verdict (Fig. A.8b). The year 2013
370 has the lowest annual medians on record, so beginning there manufactures
371 an upward trend. With 9–11 annual values and three gap years the test has
372 low power, and the correct reading is absence of evidence, not evidence of
373 stability. We recommend that trend analyses in campaign-based lake records
374 report this sensitivity as a matter of course.

375 4. Discussion

376 4.1. Implications for harmonized multi-mission archives

377 The virtual-constellation approach (Claverie et al., 2018; Pahlevan et al.,
378 2019) has become the default for multi-decadal lake monitoring, and the Roy
379 et al. (2016) coefficients are its most reused transform. Our results place
380 a boundary on where that default is safe. Those coefficients were fitted on
381 terrestrial targets and carry additive intercepts appropriate to surfaces one
382 to two orders of magnitude brighter than clear water. Over Lake Titicaca the
383 intercept exceeds the native over-water signal by a factor of 69.8 in the near-
384 infrared and by 2.5 even in blue, and the two sensors remain 99.8% separable
385 after the transform. That water reflectance from these products is unreliable
386 is already documented (Pahlevan et al., 2019; Masek et al., 2021); what our
387 paired observations add is the magnitude — the problem is not confined
388 to the near-infrared, and it is large enough to make the harmonized bands
389 function as sensor labels for a machine-learning model — and the finding that
390 local recalibration repairs the aggregate spectral signature without repairing
391 per-observation agreement in blue.

392 The practical consequence is direct. A machine-learning model trained
393 on a pooled, land-harmonized archive can achieve apparently good accuracy
394 while having learned two sensor-specific mappings joined by a readable sensor
395 label. That is not a harmonized retrieval, and it will not behave as one when
396 applied to a scene from either sensor alone. The diagnostic is cheap and we
397 recommend it as routine: train a classifier to predict the source sensor from
398 the harmonized features and report its accuracy against the majority-class
399 baseline. If the sensors remain separable, the harmonization has not done

400 its job. For clear oligotrophic lakes we further recommend deriving water-
401 specific coefficients from same-day paired observations, reporting per-band
402 agreement, and — where blue-band agreement is as poor as we find here —
403 reporting single-sensor retrievals alongside any pooled product.

404 *4.2. Implications for validation design*

405 Blocking by station is the standard remedy for spatial autocorrelation
406 in this literature, and in our data it changes nothing: random and station-
407 blocked folds give identical accuracy. Read conventionally, that would be
408 reassuring. The null models show it is not. Under campaign-blocked folds
409 the station-mean null still explains 21% of variance and the zone-mean null
410 29%, while the campaign-date-mean null explains none. In a campaign-based
411 monitoring programme the whole lake is sampled within a few days under
412 shared atmospheric and limnological conditions, so observations cluster by
413 campaign far more strongly than by location; station-blocked folds leave that
414 structure entirely intact.

415 The general point is that the blocking unit should be diagnosed, not
416 assumed. Reporting null models under each candidate design costs little
417 and identifies the unit that carries the dependence. This applies well beyond
418 Lake Titicaca: most large-lake monitoring programmes sample in campaigns,
419 so the same mismatch between the conventional blocking unit and the real
420 dependence structure is likely to be widespread.

421 For our retrieval, the honest figure is $R^2 = 0.574$ (RMSE 1.86 m) under
422 campaign-blocked validation, against 0.287 for a zone-mean null: the model
423 roughly doubles the skill available from knowing only which basin a sample
424 came from. That is a real but modest contribution, and stating it that way

425 is more useful to a manager than an unqualified R^2 .

426 4.3. Operational envelope for Lake Titicaca

427 Our RMSE of 1.86 m spans roughly 12% of the observed Z_{SD} range
428 and sits below the published range for Z_{SD} retrieval ($R^2 \approx 0.6$ – 0.9 ; Zhang
429 et al., 2022; Lee et al., 2015), obtained here under a predominantly clear
430 oligotrophic regime and a stricter validation design. It is also approaching
431 the granularity of the reference measurement itself: disk readings are recorded
432 to the nearest half metre, giving only 73 distinct values across 812 match-ups.

433 Four constraints bound operational use. The retrieval carries a 1.01 m
434 positive bias in Bahía de Puno — the most eutrophic and most vulnerable
435 zone — so mapped values there must be read with their intervals and con-
436 firmed by field sampling. Leave-one-zone-out results show the model cannot
437 extrapolate to an unsampled optical regime; integration must be incremen-
438 tal, each new campaign both validating the current map and extending the
439 training set. The retrieval is calibrated and validated for the austral dry sea-
440 son only, and the reason is worth stating precisely because it determines the
441 remedy. It is not that wet-season imagery is unusable at 3810 m, nor that our
442 match-up criteria excluded it: it is that the field programme has produced
443 a single wet-season campaign in fourteen years, and that campaign predates
444 both satellites. Wet-season blooms in Bahía de Puno — the phenomenon a
445 transparency product would most usefully warn about — are therefore en-
446 tirely unrepresented, and the model should not be applied between December
447 and March.

448 This converts the limitation into a specific, actionable recommendation
449 for the monitoring authorities. Extending the protocol to the wet season does

not require a better algorithm; it requires field data that do not yet exist. A single additional wet-season campaign per year, timed to any cloud-free window and covering the three trophic zones, would within a few years supply enough independent campaign-blocks to calibrate and validate a wet-season retrieval under the same design used here. Because the binding constraint is sampling rather than sensing, this is among the cheapest available improvements to Lake Titicaca’s water-quality monitoring capability. And because conformal coverage degrades at both extremes of the transparency range, the nominal 90% interval should be treated as optimistic in the clearest and most turbid water.

Within those bounds the product is useful to the regional monitoring authorities (IMARPE, ALT, OEFA, and the binational management framework). A calibrated ± 3 m interval resolves zone-scale contrasts and multi-metre change, which is the scale at which clarity loss in Bahía de Puno would be actionable, and a single transboundary product under one calibration supports comparable assessment across the Peru–Bolivia boundary where coordinated field monitoring is logistically difficult. It does not resolve sub-metre change and should not be presented as doing so.

4.4. *Limitations*

Beyond the constraints already stated, three limitations bound this work. The in-situ record is episodic, with no campaigns in 2020, 2021, 2023 and only one wet-season campaign in fourteen years, so the temporal hold-out rests on two test years, the trend analysis on 9–11 annual values, and the seasonal envelope on none. Transparency above approximately 15 m saturates the optical signal. And this is a single lake: the two diagnostics we propose are

475 general, but the water-specific coefficients in Table A.3 are local to Titicaca
476 and must be refitted elsewhere.

477 5. Conclusions

478 Using 1002 satellite–field match-ups over twelve years, we show that two
479 practices imported from the turbid-water remote-sensing literature fail in a
480 clear high-altitude large lake. Cross-sensor harmonization coefficients derived
481 over land leave Sentinel-2 and Landsat 99.8% separable over oligotrophic
482 water, because their additive intercepts exceed the native over-water signal
483 by up to a factor of 69.8; and validation blocked by monitoring station leaves
484 the dominant dependence — between campaigns — entirely intact.

485 Correcting both, a Random Forest retrieves Secchi depth with $R^2 = 0.574$
486 and RMSE 1.86 m, roughly double the skill of a zone-mean null, with confor-
487 mal prediction intervals that are calibrated marginally but under-cover at the
488 extremes of the transparency range. Skill collapses under extrapolation to an
489 unsampled trophic zone; chlorophyll-*a*, suspended solids and thermal tem-
490 perature are not retrievable at all; and no transparency trend is significant
491 once the full in-situ record is used rather than the satellite-era subset.

492 We therefore recommend two routine checks for multi-mission lake water-
493 quality products, neither of which requires new methodology: report the
494 accuracy of a classifier two-sample test on the harmonized features, and re-
495 port null models under each candidate validation blocking unit. Both cost
496 one model fit, and both distinguish genuine optical retrieval from sensor la-
497 belling and sampling design. For Lake Titicaca the resulting product is a
498 dry-season, zone-scale transparency indicator with honest ± 3 m intervals —

less than has sometimes been claimed for such lakes, and more than the region has had.

Appendix A. Supplementary Figures and Tables

Table A.2: Effect of the Roy et al. (2016) OLI→MSI transform over Lake Titicaca water. Native OLI is the Collection-2 surface reflectance recovered by inverting the transform. Reflectance is dimensionless.

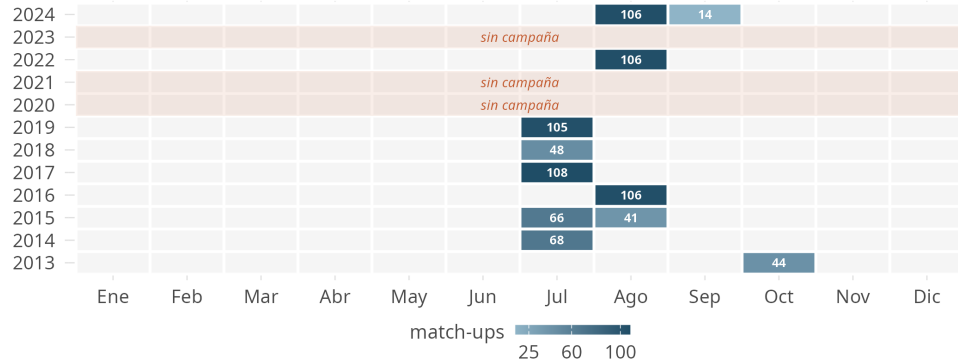
Band	Roy intercept a	Native OLI median	Intercept / signal	Harmonized LS median	Harmonized / S2 median
Blue	0.0183	+0.0072	2.5	0.0247	1.74
Green	0.0123	+0.0089	1.4	0.0206	1.61
Red	0.0123	+0.0013	10.6	0.0134	2.66
NIR	0.0448	−0.0006	69.8	0.0443	15.42
SWIR1	0.0306	+0.0013	23.2	0.0317	13.34
SWIR2	0.0116	+0.0013	8.2	0.0129	5.70

CRedit authorship contribution statement

Dina Maribel Yana-Yucra: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft. **Richar Andre Vilca-Solorzano:** Methodology, Software, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Fred Torres-Cruz:** Supervision, Resources, Writing – review & editing.

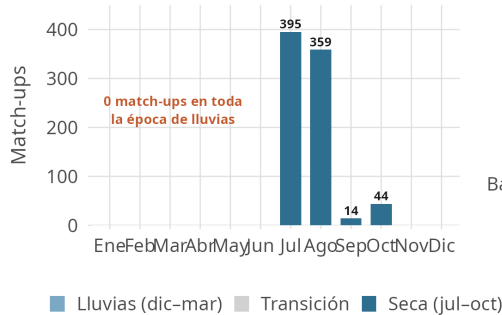
(a) Tres años sin campaña y ni un solo dato fuera de la época seca

Número de match-ups satélite-campo por año y mes. No existe dato de 2020, 2021 ni 2023, así que el hold-out «2022–2024» sólo contiene 2022 y 2024



(b) Sólo hay datos de época seca

Las floraciones de Bahía de Puno ocurren en época de lluvias, que el dataset no cubre en absoluto



(c) El gradiente trófico domina el Secchi

Sólo 73 valores distintos en 812 match-ups: las lecturas se anotan al medio metro, así que el RMSE de 1.86 m roza la granularidad de la propia medida de referencia

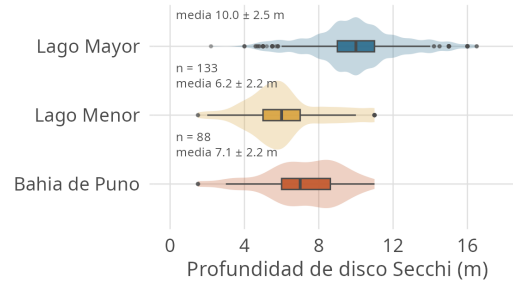


Figure A.5: What the dataset actually contains. (a) Match-ups by year and month; three years have no campaign at all, so a hold-out described as “2022–2024” contains only 2022 and 2024. (b) Seasonal coverage: zero match-ups in the wet season, when blooms occur in Bahía de Puno. (c) Secchi distribution by trophic zone; the reference measurement is quantized at roughly half a metre.

Table A.3: Water-specific LS→S2 recalibration fitted on 270 paired station–date observations, of the form $\rho_{S2} = a + b \rho_{LS, Roy}$.

Band	a	b	Pearson r	Agreement
Blue	+0.0084	0.258	0.26	unusable
Green	−0.0009	0.665	0.71	good
Red	−0.0005	0.427	0.52	weak
NIR	−0.0250	0.631	0.62	acceptable
SWIR1	−0.0223	0.770	0.64	acceptable
SWIR2	−0.0057	0.615	0.60	acceptable

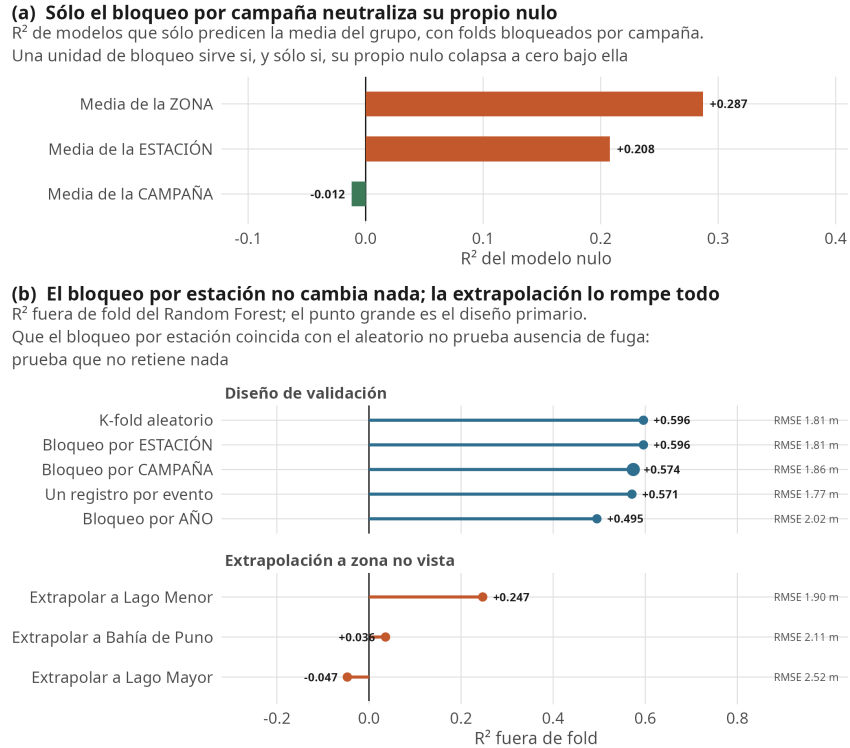
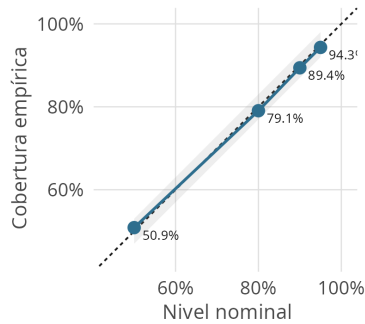


Figure A.6: Diagnosing the blocking unit. (a) Null models evaluated under campaign-blocked folds: only the campaign-mean null collapses, so only campaign blocking is doing work. (b) Out-of-fold R^2 under each validation design and under extrapolation to an unseen trophic zone. That station blocking matches the random split is not evidence of no leakage; it is evidence that station blocking retains nothing.

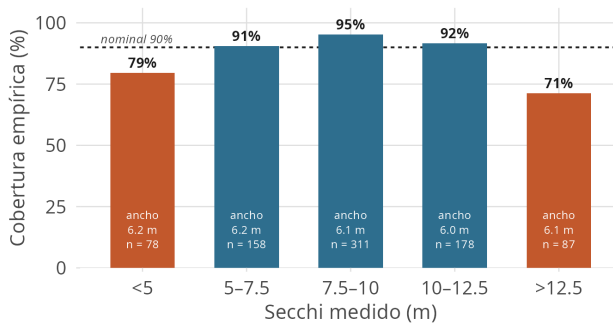
(a) Calibrados... en promedio

Cobertura empírica fuera de fold frente a la nominal. La banda gris es ± 3 puntos que la anchura es constante (~ 6.1 m) y el intervalo no puede adaptarse: infracubre en el agua más clara



(c) Pero no en los extremos del rango

El conformal split usa un único cuantil global, así a la nominal. La banda gris es ± 3 puntos que la anchura es constante (~ 6.1 m) y el intervalo no puede adaptarse: infracubre en el agua más clara



(b) El intervalo al 90% frente a la medida real

Match-ups ordenados por transparencia medida. La línea es la predicción, la banda el intervalo conformal al 90%. 86 de 812 puntos quedan fuera

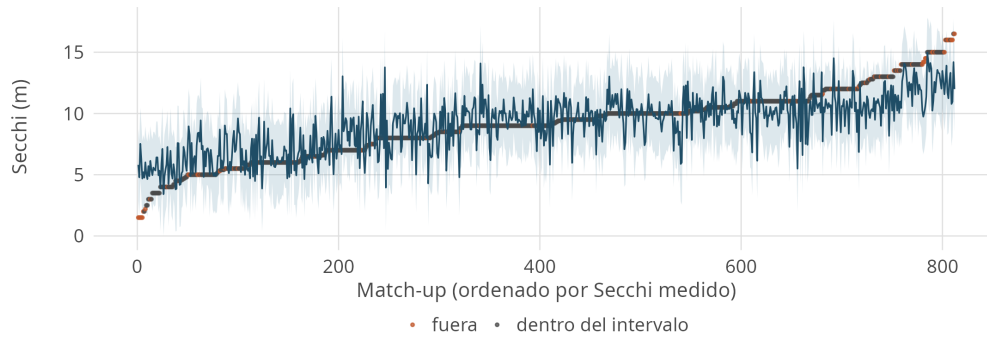
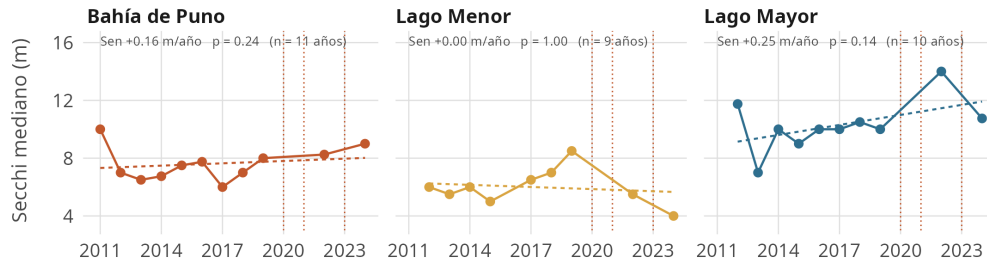


Figure A.7: Conformal prediction intervals under campaign-blocked validation. (a) Empirical versus nominal coverage: marginally calibrated across the 50–95% range. (b) The 90% interval against the measured value. (c) Coverage conditional on measured transparency. Because split conformal uses a single global residual quantile, the interval width is constant (≈ 6.12 m) and cannot adapt, so the marginal guarantee is bought by over-covering in the middle of the range and under-covering at both extremes.

(a) Ninguna zona muestra tendencia significativa en el registro completo

Mediana anual de transparencia in situ (2011–2024, 734 lecturas). Las líneas punteadas verticales marcan los años sin campaña: 2020, 2021 y 2023



(b) El veredicto lo fija el año en que empieza la serie, no el lago

Mismo test de Mann-Kendall, mismos datos, distinto año de inicio. Arrancar en 2013 —el año con las medianas más bajas del registro— convierte dos zonas en «significativamente más transparentes»

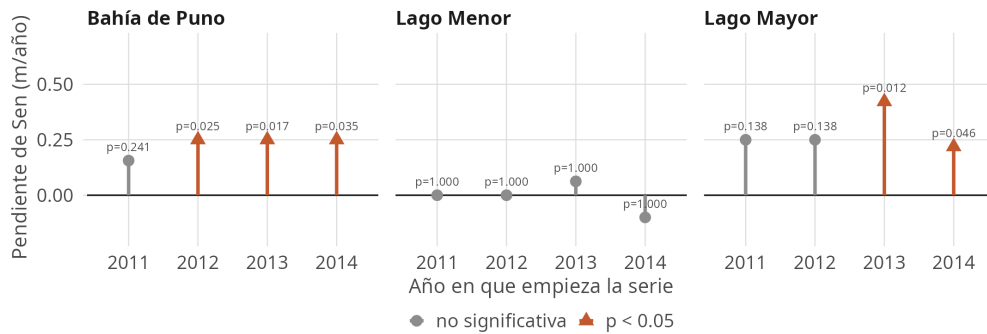


Figure A.8: Transparency trends and their fragility. (a) Annual median in-situ transparency by zone over the full 2011–2024 record, with Mann–Kendall/Sen statistics; no trend is significant. (b) The same test with different start years: beginning the series in 2013, the lowest-median year on record, turns two zones significantly positive.

Table A.4: Head-to-head benchmark, campaign-blocked GroupKFold, $n = 812$.

Algorithm	R^2	RMSE	MAE	Bias
Classical blue/green, unbounded ^a	−2.458	5.29	1.91	−0.19
Classical blue/green, bounded ^b	0.097	2.71	2.13	−0.46
Multiband linear	0.323	2.34	1.78	−0.01
Random Forest (this study)	0.574	1.86	1.44	−0.02

RMSE, MAE and bias in metres.

^a back-transformed without bounding predictions; see text.

^b standard form of Kloiber et al. (2002).

508 Declaration of competing interest

509 The authors declare that they have no known competing financial inter-
510 ests or personal relationships that could have appeared to influence the work
511 reported in this paper.

Table A.5: Model selection under nested cross-validation (Optuna inside an inner campaign-blocked split of each outer training partition). Selected configurations; the full grid is in the repository.

Model	Features	R^2	RMSE (m)
Ensemble of the four tuned models	all 12	0.583	1.84
LightGBM, tuned	all 12	0.581	1.84
XGBoost, tuned	all 12	0.577	1.85
Random Forest, tuned	all 12	0.576	1.85
Random Forest, untuned (reported)	all 12	0.574	1.86
Extremely Randomized Trees, untuned	all 12	0.573	1.86
XGBoost, tuned, $\log Z_{SD}$	all 12	0.561	1.89
LightGBM, tuned	visible only	0.544	1.92
LightGBM, tuned	ratios only	0.488	2.04
Seed-to-seed spread of the untuned Random Forest: R^2 range 0.003 over five seeds.			

Table A.6: Out-of-fold error by trophic zone, campaign-blocked validation.

Zone	R^2	RMSE (m)	Bias (m)	n
Bahía de Puno	0.221	1.90	+1.01	88
Lago Menor	0.211	1.94	+0.08	133
Lago Mayor	0.448	1.83	−0.19	591

512 **Data availability**

The satellite–field match-up datasets and the complete analysis pipeline, including the scripts that generate every number, table and figure in this article, are openly available at:

513 <https://github.com/Andre031222/harmonized-titicaca-transparency>



Repository

Raw imagery is freely available via Google Earth Engine; the in-situ measurements derive from published IMARPE/ALT and OEFA monitoring records.

514 **Acknowledgements**

515 We thank IMARPE, ALT and OEFA for the publicly available monitoring
516 records that made this study possible. This research received no external
517 funding.

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